北京郵電大學

Queen Mary University of London

For examiners' use only

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Total	

EBU5212 A

Joint Programme Examinations 2023/24

EBU5212 Communication and Networks

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Yan SUN, Dr Habiba Akter

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) **Figure 1** shows a form of Huffman tree. Determine the code efficiency of this Huffman tree.

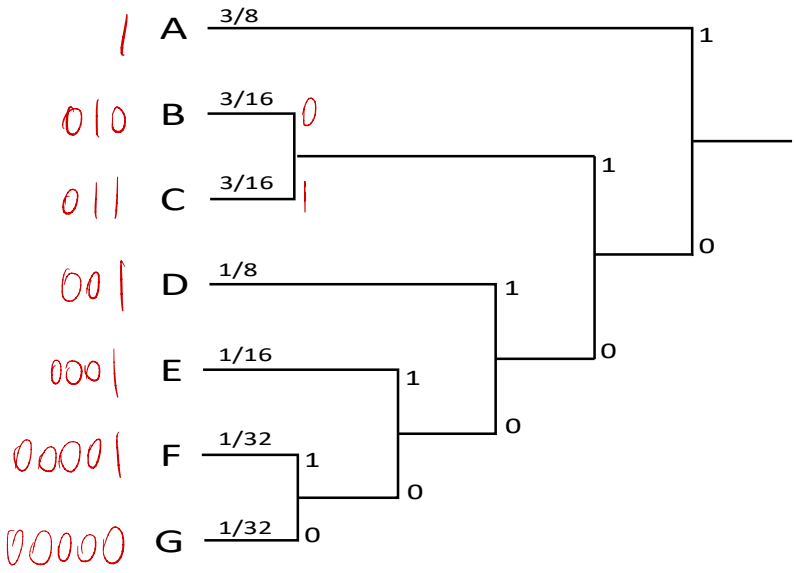


Figure 1

[12 marks]

	Do not write in this column
$\eta = \frac{H}{L}$	
$L = \frac{3}{8} + \frac{3}{16}\times 3\times 2 + \frac{1}{8}\times 3 + \frac{1}{16}\times 4 + \frac{1}{32}\times 5\times 2$	
$= \frac{3}{8} + \frac{4}{8} + \frac{3}{8} + \frac{1}{4} + \frac{5}{16}$	
$= \frac{6+18+6+4+5}{16} = \frac{39}{16} = 2.4375$ bit/symbol	
$H = \sum_{i=1}^{n} P_i \log_2\left(\frac{1}{P_i}\right)$	
$= -\left(\frac{3}{8}\log_2\frac{3}{8} + \cdots\right) = 2.125$ bits/symbol	
$\eta = \frac{H}{L} = 94.4\%$	

b) A systematic linear block code with the following parity check matrix H is employed to protect information against errors:

$$H = \left[\begin{array}{ccc|cccc} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

$H = [P^T \mid I_{n-k}]$

$G = [I_k \mid P]$

[13 marks]

i) Find all the codewords via G matrix (generator matrix).

(8 marks)

ii) Decode the following received sequence r = 1111010.

(5 marks)

	Do not write in this column
i) $P^T = \begin{bmatrix} 1&1&0 \\ 0&1&1 \\ 1&1&1 \\ 0&1&0 \end{bmatrix}$ $P = \begin{bmatrix} 1&0&1&0 \\ 1&1&1&1 \\ 0&1&1&0 \end{bmatrix}$	
$\therefore G = \begin{bmatrix} 1&0&0&1&0&1&0 \\ 0&1&0&1&1&1&1 \\ 0&0&1&0&1&1&0 \end{bmatrix}$ $k=3$ $n=7$ $m=4$	
x_0 000 C_0 0000000	
x_1 001 C_1 0010110	
x_2 010 C_2 0101111	
x_3 011 C_3 0111001	
x_4 100 C_4 1001010	
x_5 101 C_5 1011100	
x_6 110 C_6 1100101	
x_7 111 C_7 1110011	
ii) $S = rH^T = [1001] = (c+e)H^T = eH^T$ $r = [1111010]$ $H^T = \begin{bmatrix} 1&0&1&0 \\ 1&1&1&1 \\ 0&1&1&0 \\ 1&0&0&0 \\ 0&1&0&0 \\ 0&0&1&0 \\ 0&0&0&1 \end{bmatrix}$	

$eH^T = [1001]$

e_0	0000000	0000
e_1	1000000	1010
e_2	0100000	1111
e_3	0010000	0110
e_4	0001000	1000
e_5	0000100	0100
e_6	0000010	0010
e_7	0000001	0001

$H^T = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$(e_4+e_7)H^T = (e_2+e_3)H^T$

$1111010 = r$

$c = 1110011 \;/\; 1001010$

$c_7 \qquad c_4$

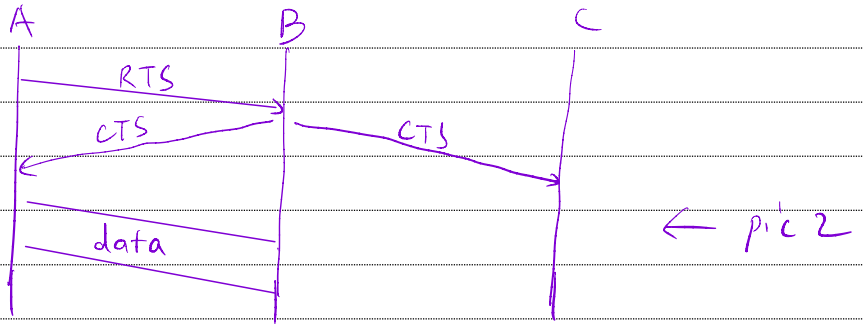
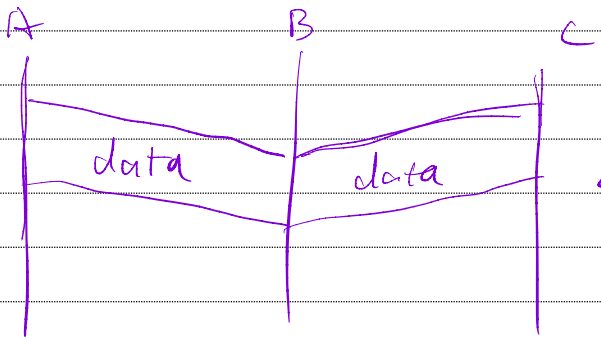
	13 marks

Question marking: $\frac{}{12} + \frac{}{13} = \frac{}{25}$

Question 2

a) What is the "hidden terminal" problem? Provide your answer with the aid of illustration.

[7 marks]

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 ← pic 2 hidden terminal problem occurs in wireless communication, when two nodes A and C are out of each other's range, but both communicate with node B. This leads a collision at the shared node when A and C transmit data at the same time. (like pic 1)  ← pic 1 Sol: use RTS (request to send) + CTS (clear to send) A sends RTS to B, B responds CTS and broadcast. (pic 2)	
	7 marks

b) Describe how host computer 192.168.1.2 is able to deliver Internet Protocol datagrams to host 192.168.1.4 on the same Ethernet network, via bridge BR, illustrating your answer with reference to **Figure 2a**. Your answer should include a full description of the role played by the Address Resolution Protocol (ARP) and assume that no previous communication has taken place for some time.

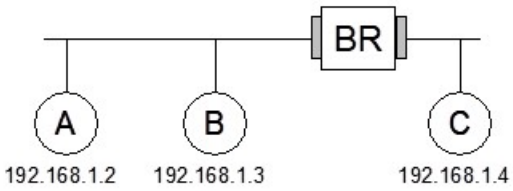


Figure 2a

[10 marks]

	Do not write in this column
① Initial state: Host 192.168.1.2 knows the destination IP but it doesn't know the exactly MAC address. 192.168.1.2 checks its ARP cache for the MAC address, it's empty, so ARP is triggered.	
② 192.168.1.2 sends a broadcast ARP request on the network. ARP packet: A' MAC / 192.168.1.2 / FF:FF:FF:FF:FF:FF / 192.168.1.4	
③ broadcast the ARP frame, go through the BR to the C. C gets ARP request, and then send a response to A. C' MAC / 192.168.1.4 / A' MAC / 192.168.1.2	
④ A gets the ARP response and update its ARP table.	
⑤ A sends its data by creating an Ethernet frame: Preamble | SFD | C'MAC | A's MAC | Type | Data | CRC	
⑥ C receive the upper frame and check the address, decapsulate the IP	
	10 marks

c) The bridge is now replaced with a router RT, as shown in **Figure 2b**. Explain any differences that would arise in the process of how host 192.168.1.2 delivers data to host 192.168.2.4. Once again, assume that no previous communication has taken place for some time and the subnet mask is /24.

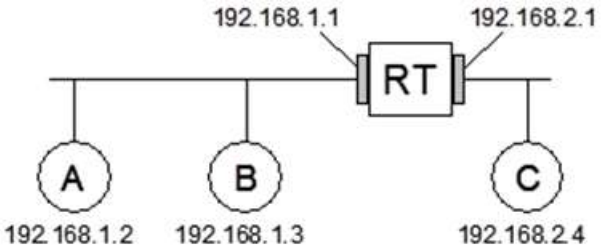


Figure 2b

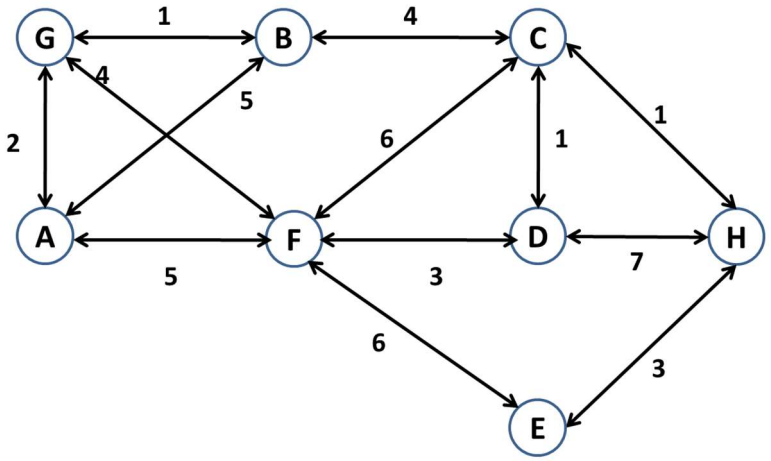
[8 marks]

	Do not write in this column
difference : in b) A.C in the same subnet, however c), A.C in different subnet,	
so A should firstly forwards the packet to the default gateway, 192.168.1.1 (router for subnet 1)	
① A checks ARP cache (empty), send ARP request for MAC of 192.168.1.1, RT response with MAC address for subnet 1.	
② A sends packet to RT (192.168.1.1) by Ethernet frame	
③ RT processing, RT examines the IP packet, checks its Routing table and find out it belongs to subnet 2	
④ RT queries for MAC of 192.168.2.4 from 192.168.2.1	
⑤ forward packet to 192.168.2.4, RT create new frame here	
⑥ receive by C	
	8 marks

Question marking: $\frac{}{7} + \frac{}{10} + \frac{}{8} = \frac{}{25}$

Question 3

a) Describe how would you determine the least-cost tree for the topology below from node D using Dijkstra's algorithm. You should also illustrate the final tree topology.



[12 marks]

			Do not write in this column
Candidate	Cost to Root	Tree	
D→C 1	(1)	D→D 0	
D→F 3	3		
D→H 7	7		
D→F 3	3	D→D 0	
D→H 7	7	D→C 1	
C→B 4	5		
~~C→F 6~~	~~7~~		
C→H 1	(2)		
D→F 3	(3)	D→D 0	
C→B 4	5	D→C 1	
H→E 3	5	C→H 1	
C→B 4	(5)	D→D 0	
H→E 3	5	D→C 1	
F→G 4	7	C→H 1	
F→A 5	8	D→F 3	
~~F→E 6~~	~~9~~		
H→E 3	(5)	D→D 0 C→H 1	
~~F→G 4~~	~~7~~	D→C 1 D→F 3	

F → A 5 8 C → B 4 H → E 5

B → G 1 6

~~B → A 5 10~~

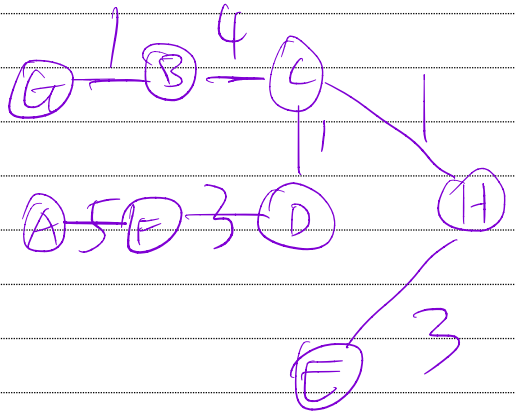
F → A 5 8 D → D 0 C → H 1 C → B 4

B → G 1 6 D → C 1 D → F 3 H → E 3

F → A 5 8 D → D 0 C → H 1 C → B 4 B → G 1

G → A 2 8 D → C 1 D → F 3 H → E 3

D → D 0
D → C 1
C → H 1
D → F 3
C → B 4
H → E 3
B → G 1
F → A 5



12 marks

b) Applying the Distance Vector Algorithms to the topology in a), produce the routing table at node F.

[7 marks]

Do not write in this column

① F's neighbours

node	cost
A	5
G	4
C	6
D	3
E	6

② routing table that neighbour send to F

~~to~~ from	A 5	G 4	C 6	D 3	E 6
A	0	2	7	8	11
B	3	1	4	5	8
C	7	5	0	1	4
D	8	6	1	0	5
E	11	9	4	5	0
F	5	4	4	3	6
G	2	0	5	6	9
H	8	6	1	2	3

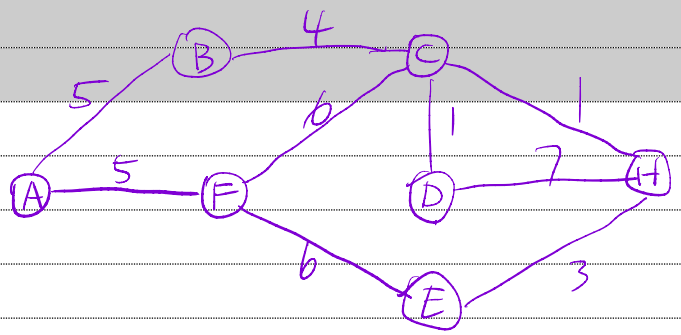
③ cost from F to other nodes

Dest	Dis	first hop
A	5	A
B	5	G
C	4	D
D	3	D
E	6	E
F	0	-
G	4	G
H	5	D

7 marks

c) Produce the updated routing table at node F with Distance Vector algorithm when node D is disconnected with node F and node G moves away from the topology in b).

[6 marks]

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① F's neighbour:	

Nei	Cost
A	5
C	6
E	6

② routing table from neighbors of F

to \ from	A 5	C 6	E 6
A	0	9	11
B	5	4	8
C	9	0	4
D	10	1	5
E	11	4	0
F	5	6	6
H	10	1	3

③ F to others

Destination	Cost	Next hop
A	5	A
B	10	A
C	6	C
D	7	C
E	6	E
F	0	—
H	7	C

	6 marks

Question marking: $\frac{\quad}{12} + \frac{\quad}{7} + \frac{\quad}{6} = \frac{\quad}{25}$

Question 4

a) Considering the traditional TCP in wired IP network,

[14 marks]

i) What are the advantages and disadvantages of the fact that TCP acknowledgements are cumulative?

(9 marks)

ii) What can be deduced from the reception of duplicate acknowledgements?

(2 marks)

iii) How is this information used in TCP Reno to maintain channel utilisation?

(3 marks)

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i) ad:	
① easy to generate: receiver just send 1 ACK and can tell sender that all data before that SN is successful received.	
② tolerance ACK loss: if some ACK lost, after later ACK is arrived, it can get to know which data to be retransmitted.	
③ efficiency: reduce burden	
dis:	
① delayed detection of loss: if a seg is lost, sender may not detect it immediately cause ACKs continue to	
② transmiter only knows last contiguous number of data, doesn't know which specific data to collect.	
③ transmiter may retransmit right data	
④ hard to change window size in time.	

ii)

① detection of data loss : indicate that a segment has been lost

② may occur congestion

iii)

① change threshold : set new threshold = $\frac{cwnd}{2}$

② change Cwnd : set cwnd = threshold

③ go into AI (addictive increase) mode

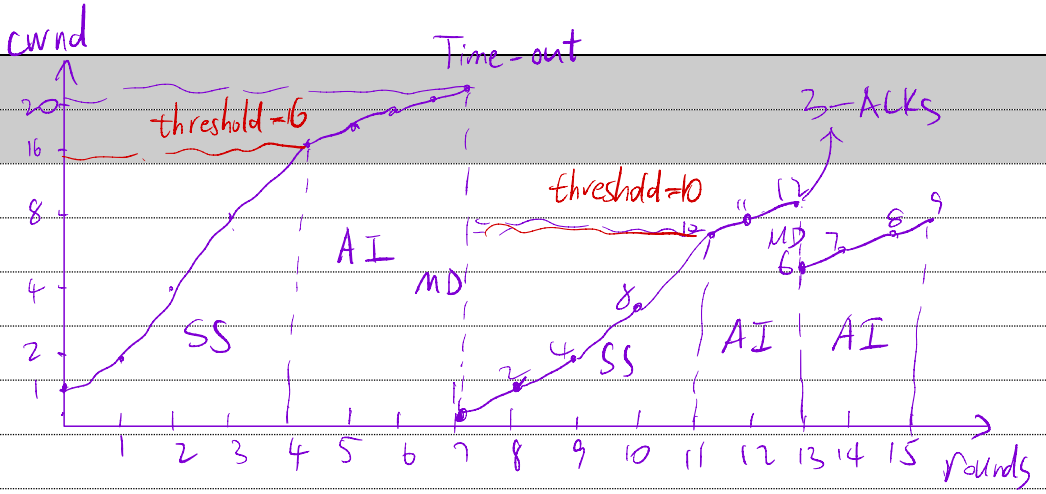
④ effective window = min (advertised window, congestion window)

14 marks

b) TCP congestion control uses 4 basic mechanisms known as Additive Increase (AI), Multiplicative Decrease (MD), Slow Start and Fast Recovery. Briefly explain each of those 4 basic mechanisms with figure illustration.

[11 marks]

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AI: after cwnd reaches threshold, run congestion avoidance algorithm, then cwnd size will addictively increase.

MD: If cwnd reaches time-out or transmitter gets 3 duplicate ACKs, it refers to congestion, cwnd will do the multiplicatively decrease, for TCP Tahoe, the window size will set to 1, for TCP RENO, set threshold = cwnd max/2, cwnd = threshold

SS: when cwnd< threshold, cwnd will increase exponentially until cwnd = threshold.

Fast Recovery: for the TCP RENO, transmiter gets 3 duplicate ACKs, transmiter will retransmit corresponding segment at once and set cwnd = $\frac{\text{cwnd max}}{2}$ = threshold and start AI.

	11 marks

Question marking: $\frac{}{14} + \frac{}{11} = \frac{}{25}$

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2023-2024

Rough Working

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2023-2024 Rough Working

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2023-2024 Rough Working